Data Science Project- Phase 2 Logbook

Project: Sentiment Analysis of Flu Vaccination Opinions

Phase: Data Cleaning and Preprocessing + EDA

Data Cleaning Entries (1-3)

ENTRY 1

Date: Feb 25- Mar 2, 2026

Dataset: YouTube Comments (JSONL)

Data Inspection

The raw YouTube comments dataset was loaded from JSONL format into a pandas DataFrame and inspected for missing values and duplicate records. The dataset schema was reviewed to confirm key fields (e.g., commentId, timestamps, and comment text) were available for preprocessing.

Cleaning & Standardization

Duplicate comments were removed using the unique commentId, and empty/invalid text entries were filtered out. Basic text cleaning was applied to normalize whitespace and remove embedded URLs while keeping the original raw text unchanged for traceability. Timestamp fields were also parsed into datetime format to ensure consistent handling in later analysis.

Advanced Text Preprocessing

Based on instructor guidance, advanced preprocessing was applied to prepare the text for analysis. This included lowercasing, punctuation removal, tokenization, stopword removal, and lemmatization (spaCy). Both a cleaned text field and a fully processed text field were retained to support transparency and reproducibility.

Vectorization & Saved Outputs

TF-IDF vectorization (unigrams and bigrams) was applied to the processed text as part of preprocessing. The TF-IDF sparse matrix was saved in .npz format along with the fitted vectorizer, the feature list, and a row-mapping file linking vectors back to commentId. A small frequency preview was included only as a preprocessing validation step (pre-EDA) to confirm preprocessing correctness. The final structured dataset was exported as a cleaned CSV file for Phase 2.

ENTRY 2

Date: Feb 25- Mar 2, 2026

Dataset: GNews Articles

Data Inspection

The raw GNews dataset was loaded from JSONL format into a pandas DataFrame and inspected for missing values and duplicate records. The schema was reviewed, and nested fields (specifically the source dictionary) were identified for normalization.

Schema Normalization & Cleaning

The source dictionary was flattened by extracting source_name and source_url into separate columns. A unified text field was constructed by combining the article title, description, and content. Duplicate entries were removed using url and/or id, and empty or invalid text records were filtered out. Publication timestamps were converted to datetime format to ensure consistency.

Advanced Text Preprocessing

Advanced text preprocessing was applied to the cleaned article text, including lowercasing, punctuation removal, tokenization, stopword removal, and lemmatization using NLTK and spaCy. The processed text field was created to support consistent downstream analysis.

Vectorization & Saved Outputs

TF-IDF vectorization (unigrams and bigrams) was applied to the processed text as part of preprocessing. The resulting sparse TF-IDF matrix was saved in .npz format, and the fitted vectorizer was saved in .pkl format to ensure reproducibility of the feature space. The final cleaned structured dataset was exported as a CSV file for Phase 2.

A brief raw frequency preview was included solely as a preprocessing validation step (pre-EDA) to confirm the correctness of text transformation.

ENTRY 3

Date: Feb 25 - Mar 2, 2026

Dataset: openFDA – Influenza Adverse Event Reports

Data Inspection

The raw openFDA influenza adverse event dataset (JSON format) was loaded into Google Colab using pandas. Initial inspection showed 1,000 records and 27 columns, with no duplicate report IDs.

Several missing values were detected, and the patient field was identified as a nested dictionary requiring restructuring.

Data Restructuring

The nested patient field was flattened using json_normalize, increasing the dataset to 34 columns. Because each report contained a list of adverse reactions, the patient_reaction field was expanded using explode(), transforming the dataset into 7,566 reaction-level rows. Reaction dictionaries were further flattened to extract the standardized reaction term (reaction_reactionmeddrapt) to enable accurate frequency analysis.

Cleaning & Standardization

Text fields were normalized (lowercasing and trimming), invalid placeholders were converted to missing values, and date columns were converted to datetime format using the correct format (YYYYMMDD). Patient age was converted to numeric, and unrealistic values (<0 or >120) were removed to ensure data validity and analytical reliability. Patient sex was retained in its original numeric coding (1 = Male, 2 = Female).

Text Vectorization

As part of preprocessing, the cleaned reaction column was transformed into numerical format using CountVectorizer (Bag-of-Words). The number of features was limited to 500 (max_features=500) to control dimensionality and reduce file size while preserving the most frequent and meaningful terms. Each unique word became a feature, and each reaction entry was represented as a frequency-based vector. The resulting matrix had shape (7,566 × 500). The most frequent terms included pain, decreased, increased, blood, drug, disorder, infection, cough, dyspnoea, and influenza.

Final Output & Rationale

The final structured dataset contained 7,566 rows and was saved as openfda_influenza_clean.csv. The vectorized representation was saved separately as openfda_vectorized_bow_500.csv. The restructuring process was necessary due to nested JSON structures and list-based reaction fields, which would otherwise prevent meaningful exploratory analysis. Bag-of-Words was selected because reaction terms are short and structured, making term-frequency representation appropriate for Phase 2 preprocessing requirements.

Exploratory Data Analysis Entries (EDA) (5-9)

ENTRY 4

Secondary Dataset Acquisition (Zindi/Kaggle)

Date: Mar 3, 2026

Task: Selected and obtained a secondary text dataset related to vaccination sentiment for Phase 2 analysis

Result: Downloaded the Zindi/Kaggle dataset "*Sentiment towards Vaccination*" (training set) as Train.csv, containing 10,001 labeled tweets with fields including tweet_id, safe_text, label, and agreement.

Link:
https://www.kaggle.com/datasets/alfredkondoro/to-vaccinate-or-not-to-vaccinate-zindi?resource=download&select=Train.csv

Challenge: The dataset was not flu-specific, and metadata about time/location context was limited in the dataset description

Solution: Adopted the dataset as a secondary benchmark for general vaccination sentiment due to its labeled tweet text format and manageable size; documented the limitation that it may represent broader vaccine discourse rather than flu-only discussion.

ENTRY 5

Secondary Dataset Cleaning & EDA (Zindi Tweets)

Date: Mar 3, 2026

Phase: Exploratory Data Analysis (EDA)

Task: Performed cleaning, preprocessing, and text-based EDA on the Zindi tweet dataset in Google Colab to support Phase 2 secondary data analysis

Result: Cleaned the dataset (validated labels, removed invalid entries), generated EDA summaries and visualizations, and exported a structured cleaned dataset (zindi_cleaned_dataset.csv) along with TF-IDF artifacts for reproducibility.

Data Cleaning & Validation

The dataset was loaded and inspected for missing values, duplicates, and label consistency. One anomalous label value (0.666667) was detected and removed to ensure the dataset contained only valid sentiment classes (-1 negative, 0 neutral, 1 positive). Empty/whitespace-only text entries were filtered out, and labels were mapped to readable categories for analysis.

Text Normalization & Stopword Handling

Tweets were normalized (lowercasing, URL/punctuation cleanup) to create a comparable text column for EDA (text_norm). Common English stopwords were removed from frequency-based analyses, and dataset-specific placeholders such as user, url, and amp (HTML artifact from "&") were excluded to avoid biasing word counts and word clouds with non-informative tokens.

EDA Techniques & Visualizations

EDA was performed using statistical summaries and plots, including sentiment distribution charts, word/character length analysis, unigram and bigram frequency analysis, word clouds by sentiment class, and topic modeling (LDA) to identify prominent themes. Correlation and numeric relationship checks were applied using features such as text length metrics and the annotation agreement score.

Challenges & Handling

During outlier review, one extreme-length tweet contained inappropriate language. To keep the notebook suitable for submission, the tweet text was redacted in printed outputs while preserving the underlying data for reproducibility.

Outputs Saved

The final cleaned structured dataset was exported as zindi_cleaned_dataset.csv. TF-IDF vectorization was performed (unigrams + bigrams), and outputs were saved in a lightweight reproducible format (sparse matrix .npz + vectorizer file) for use in later analysis stages.

<u>ENTRY 6</u>

Date: Mar 4, 2026

Dataset: openFDA – Influenza Adverse Event Reports

Phase: Exploratory Data Analysis (EDA)

Text Exploration

To further explore the reaction field, unigram and bigram analysis were applied using CountVectorizer. Unigrams identified the most frequent individual medical terms, while bigrams captured short multi-word patterns that may represent more specific adverse event expressions. The results were consistent with earlier frequency analysis and confirmed the dominance of influenza-related and respiratory symptoms.

Word Cloud Visualization

A word cloud was generated using all reaction terms combined into a single corpus. This visualization provided a qualitative overview of dominant medical terms and supported the numerical frequency

findings. Additional segmented word clouds were generated by sex and age group to visually compare patterns across demographic categories.

Demographic Cross Analysis

Patient age was grouped into meaningful categories (0–17, 18–40, 41–65, 65+) to improve interpretability. Reaction distributions were analyzed across these age groups using cross-tabulation and visualized through bar charts and heatmap-style comparisons.

Reaction frequencies were also compared across sex categories (Male and Female) to identify potential differences in reporting patterns.

Bivariate Relationship Analysis

The dataset was aggregated at the report level to calculate the number of reactions per safety report. A scatter plot was generated to examine the relationship between patient age and the number of reactions reported. Correlation analysis (Pearson and Spearman) indicated a weak association, suggesting that reaction volume per report does not strongly depend on patient age.

Missing Data Assessment

A detailed missing value summary was calculated, including both counts and percentages. Missing values were primarily concentrated in patient_age (~18%) and patient_sex (~5%). No significant structural issues were detected in the core analytical variables.

EDA Summary

The extended analysis provided deeper insights into text patterns, demographic distributions, and variable relationships within the openFDA influenza dataset. These findings enhance interpretability and strengthen the dataset's suitability for cross-dataset comparison in the final project phase.

ENTRY 7

Date: Mar 4, 2026

Dataset: YouTube Comments – Flu Vaccination Opinions

Phase: Exploratory Data Analysis (EDA)

Temporal Trend Analysis

The number of comments was aggregated monthly based on the publication timestamp to examine how discussion activity changed over time. The visualization shows that comment activity was relatively low in earlier years but gradually increased, with noticeable growth after 2020. This trend indicates increasing public engagement and discussion surrounding influenza vaccination topics on YouTube.

Comment Length Distribution

Comment length was calculated based on the number of characters in each comment and visualized using a histogram. The results show that most comments are relatively short, which is typical for social media platforms. However, a smaller portion of comments are longer, suggesting that some users provide more detailed opinions or personal experiences.

Word Frequency Analysis

To identify common discussion topics, the most frequent words were extracted using **CountVectorizer** after text preprocessing. Frequently appearing words included *flu, vaccine, shot, sick,* and *covid*. These results indicate that discussions mainly focus on influenza vaccination, illness experiences, and related health topics.

Bigram Analysis

Bigram analysis was applied using **CountVectorizer with ngram_range=(2,2)** to identify common two-word expressions. The most frequent bigrams included *flu shot, got flu, flu vaccine, flu shots,* and *getting flu*. These patterns suggest that many users discuss vaccination experiences or flu-related health events.

Sentiment Analysis

Sentiment analysis was conducted using the **VADER sentiment analyzer**, which classified comments into positive, negative, and neutral categories. The results show that negative comments appear most frequently, followed by positive comments, while neutral comments represent a smaller portion. This indicates that discussions include both supportive opinions and critical or negative experiences.

Sentiment Score Distribution

The compound sentiment scores produced by VADER were analyzed to examine the overall polarity of comments. Scores range from −1 (very negative) to +1 (very positive). The distribution shows many values clustered around zero, indicating a mixture of neutral opinions alongside positive and negative reactions.

Word Cloud by Sentiment

Word clouds were generated separately for positive, neutral, and negative comments using the processed text. Each group of comments was combined into a single text corpus and visualized using the WordCloud library. This visualization helps compare the most frequent words across different sentiment categories.

Topic Modeling

Topic modeling was applied using **Latent Dirichlet Allocation (LDA)** to identify underlying discussion themes in the comments. Several topics were extracted from the processed text, representing recurring themes such as vaccination experiences, COVID-19 references, post-vaccination reactions, and discussions related to medical trust and research.

Topic Distribution

The dominant topic for each comment was identified and visualized using a bar chart. The distribution shows that certain topics appear more frequently than others, highlighting the main themes discussed in the dataset.

Feature Correlation Analysis

Correlation analysis was performed between several numerical features, including **word count, text length, sentiment score, and like count**. A correlation heatmap showed a strong positive relationship between word count and text length, while other relationships were relatively weak.

Sentiment vs Comment Length

A scatter plot was created to examine the relationship between comment length (word count) and sentiment score. The results show that both positive and negative sentiments appear across different comment lengths, indicating no strong relationship between comment length and sentiment.

EDA Summary

The exploratory analysis revealed several patterns in YouTube discussions about influenza vaccination. Comment activity increased over time, while most comments remained relatively short. Word frequency and bigram analyses highlighted vaccination-related topics, and sentiment analysis revealed a mixture of positive, negative, and neutral opinions. Additional analyses such as sentiment-based word clouds, topic modeling, and correlation analysis provided deeper insights into discussion themes and comment characteristics.

ENTRY 8

Date: Mar 4, 2026
Dataset: GNews Articles Dataset
Phase: Exploratory Data Analysis (EDA)

Dataset Overview

The cleaned GNews dataset was analyzed to understand the structure and patterns within the news articles related to influenza vaccination. Basic statistical summaries were generated using pandas functions such as `df.describe()` to examine the distribution of numerical features including article length.

Article Length Analysis

Descriptive statistics were calculated for the article length feature. The results showed that the dataset contains **110 articles**, with an average article length of approximately **518 characters**. The shortest

article contained **398 characters**, while the longest article reached **1530 characters**. The distribution indicates that most articles fall within a moderate length range, with only a few longer outliers.

Title Word Count Distribution

The number of words in article titles was calculated and visualized using a histogram. The analysis showed that most titles contain a moderate number of words, typically between **8 and 14 words**, which is common for news headlines designed to be concise yet informative.

Word Frequency Analysis

Word frequency analysis was performed using `CountVectorizer` after preprocessing the article text. The most frequent words were visualized using bar charts and a word cloud. The results highlighted common terms such as **flu**, **vaccine**, **shot**, and **health**, indicating that the dataset strongly focuses on influenza vaccination discussions.

Source Distribution

A bar chart was generated to analyze the distribution of news sources within the dataset. The visualization showed that some news sources contribute more articles than others, suggesting that certain media outlets report more frequently on vaccination-related topics.

Article Length Visualization

A histogram was generated to visualize the distribution of article lengths. The plot showed that most articles fall within a similar range, while a small number of articles are significantly longer than the majority.

Outlier Detection

A boxplot was used to detect potential outliers in article length. The visualization confirmed the presence of a few unusually long articles compared to the overall distribution.

Temporal Trend Analysis

A time-based visualization was generated to observe how article publication changes over time. The analysis revealed fluctuations in the number of articles published, indicating varying levels of media attention toward influenza vaccination topics across different periods.

EDA Summary

The exploratory analysis of the GNews dataset revealed several patterns. Most articles have moderate length and concise titles, while word frequency analysis confirms that influenza vaccination is the dominant topic. The presence of a few outlier articles suggests variability in reporting depth, and the temporal analysis indicates that media attention to vaccination topics changes over time.

ENTRY 9

Date: Mar 5, 2026
Task: Report Writing and Documentation

Report Organization

During this stage, the main task focused on preparing and organizing the written report for Phase 2 of the project. The report was structured according to the project requirements and included several key sections such as the dataset overview, exploratory data analysis (EDA) results, visualizations, statistical comparisons, and summary of insights.

Documentation of Dataset Information

Detailed descriptions of the datasets used in the project were written and organized. This included explaining the structure and attributes of the primary datasets (GNews, YouTube, and OpenFDA) as well as the secondary dataset used for comparison. The documentation clarified the types of variables contained in each dataset and the role of each dataset in the analysis process.

Integration of EDA Results

The results generated from the exploratory data analysis notebooks were carefully reviewed and documented in the report. Visualizations such as histograms, bar charts, and word frequency plots were incorporated into the report to clearly present the patterns identified in each dataset. Descriptions were written to explain the meaning of each figure and how it relates to the research question.

Statistical Summary and Tables

Key statistical measures were extracted from the EDA outputs and summarized in tables within the report. These included metrics such as dataset size, mean, median, and the main numerical variables used in the analysis. The tables were designed to present the information clearly and allow easy comparison across datasets.

Cross-Dataset Comparison

A comparison section was written to analyze the similarities and differences between the primary datasets (GNews, YouTube, and OpenFDA) and the secondary dataset (Zindi). This comparison focused on aspects such as data type, dataset size, statistical characteristics, and the type of insights each dataset provides. The goal was to highlight how different data sources contribute complementary perspectives to the overall analysis.

Interpretation of Findings

Interpretations were written to explain the patterns observed in the data. For example, social media datasets reflected public discussions and opinions, while the medical dataset (OpenFDA) provided structured clinical information related to adverse reactions. These interpretations helped connect the numerical results with meaningful real-world insights.

Final Insights and Hypotheses

The final section of the report summarized the most important insights obtained from the exploratory analysis. Several hypotheses were proposed regarding how public discussions, media coverage, and clinical reports may differ in their representation of vaccination-related topics. These conclusions emphasized the importance of combining multiple data sources to gain a broader understanding of public health discussions.